University College London

Net Survival Analysis: Parametric Models

Lecture II: Parametric survival analysis in the relative survival framework

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Lecture aims

- Introduce the relative survival framework.
- Introduce a general hazard structure.
- Present an application in cancer epidemiology.
- Discuss a challenge in the relative survival framework.

The relative survival framework

Cancer epidemiology

- We are exposed to many forces of mortality: ageing, illnesses, natural disasters, accidents, crime, COVID-19, and so on.
- Cancer is a strong force of mortality that affects large groups of individuals in all countries.
- Cancer patients, the population of interest in cancer epidemiology, are exposed to the additional force of mortality due to cancer.
- The aim is to quantify the survival of cancer patients in order to compare cancer management across countries. For this, we need comparable quantities.

The typical data set

- A sample of **times to event**, possibly right-censored, $(t_1, \ldots, t_n)$, from a group of individuals.
- The vital status (or **censoring indicators**) $(\delta_1, \ldots, \delta_n)$: $\delta_i = 1$ for death, $\delta_i = 0$ for right-censored (alive). Censoring may be due to random drop-out, loss to follow-up, or administrative censoring.
- In some cases, we also know some characteristics of the individuals: **covariates** $\mathbf{x}_i = (x_{i1}, \ldots, x_{ip})^\top$, such as age, sex, deprivation level, comorbidities or tumour stage.

Survival analysis frameworks

There are three approaches to analysing survival data:

1. **Overall survival:** all-cause mortality is analysed. Not useful for comparisons between populations, as they are exposed to different causes of mortality.
2. **Cause-specific:** the cause of death is known. However, death certificates are highly unreliable in all countries.
3. **Relative survival:** we separate the hazard associated with other causes from the hazard associated with cancer.

Relative survival framework

- We would like to quantify the mortality hazard due to the cancer under study, and link it to the patients' characteristics.
- In the relative survival framework, we assume that the expected mortality hazard (from causes other than cancer) can be obtained from the general population life tables.

Excess mortality hazard regression models

- More specifically, we decompose the individual observed hazard, $h_o(\cdot;\cdot)$, as

$$h_o(t;\mathbf{x}) = h_P(A+t;\mathbf{z}) + h_E(t;\mathbf{x}),$$

where A is the age at diagnosis, $h_P(A+t;\mathbf{z})$ is the population hazard, obtained from the life tables ($\mathbf{z} \subset \mathbf{x}$: age, sex, deprivation, ...), and $h_E(t;\mathbf{x})$ is the excess hazard.

Assumptions of the relative survival framework

- The general population hazard correctly reflects the other-causes hazard in our population of interest.
- The excess hazard is interpreted as the hazard due to the cancer under study.

Background mortality

- To calculate $h_P(A+t;\mathbf{z})$, we need the life tables produced by the country where the patients live. These are publicly available.
- Depending on the country, life tables are defined by different characteristics $\mathbf{z}$; typically age, sex and year.
- In the UK, life tables are also available by deprivation level (1–5).
- Example: a patient with characteristics
  $\mathbf{z}_i = (\text{age} = 70,\ \text{sex} = \text{male},\ \text{deprivation} = 3,\ \text{year of diagnosis} = 2010)$ and
  $t_i = 1$ year $\longrightarrow$ 2011 life table $\longrightarrow$ extract the corresponding mortality rate.

Net survival

- The main quantity of interest is the **net survival**, the survival function associated with the excess hazard:

$$S_N(t;\mathbf{x}) = \exp\left\{-\int_0^t h_E(s;\mathbf{x})\,ds\right\}.$$

- It is not affected by differences in population mortality, so it is useful for comparing cancer management between countries or periods.
- Policy-making is often based on the **population net survival**, $S_N(t) = \frac{1}{n}\sum_{i=1}^{n} S_N(t;\mathbf{x}_i)$.

Net survival in the news



One in five NI cancer diagnoses via emergency routes

15 January 2020 Share

The research found a higher proportion of emergency presentations in deprived areas and among older people.

A fifth of cancer patients in Northern Ireland received their diagnosis through an emergency route, according to a major research report.

Of more than 45,000 people diagnosed from 2012 to 2016, one-fifth were diagnosed this way.

They had what's described as a "poor net survival" at three years of 23%.

The report, 'Pathways to Cancer Diagnosis', was compiled by Queen's University and the Health and Social Care Business Organisation.

- BBC News, 15 January 2020: one in five cancer diagnoses in Northern Ireland happen via emergency routes.
- Those patients had a "poor net survival" at three years of 23%.
- Net survival is the quantity routinely reported in population-based cancer research and in policy documents.

The general hazard structure

Which parametric excess hazard model?

- The excess hazard is typically modelled as

$$h_E(t; \mathbf{x}) = h_0(t) \exp\left(\mathbf{x}^\top \boldsymbol{\beta}\right).$$

- **Fully parametric** models are of interest, since they allow for prediction beyond the follow-up period, and they are typically more parsimonious.
- The main criticisms are: (i) the PH assumption does not account for time-varying effects, and (ii) the distributions used to model the baseline hazard can only capture specific shapes.

The proposed model

- Aims (Rubio et al. 2019): (i) consider more general alternatives to the PH structure; (ii) use a parametric baseline that can capture the basic shapes of interest.
- For (i), to go beyond proportional hazards, we consider the **general hazard (GH)** structure (Chen and Jewell 2001):

$$h_{\mathsf{E}}^{\mathsf{GH}}(t;\mathbf{x}_i) = h_0\left(t\exp(\mathbf{x}_i^\top\boldsymbol{\beta}_1)\right)\exp(\mathbf{x}_i^\top\boldsymbol{\beta}_2),$$
$$H_{\mathsf{E}}^{\mathsf{GH}}(t;\mathbf{x}_i) \overset{\text{(exercise)}}{=} H_0\left(t\exp(\mathbf{x}_i^\top\boldsymbol{\beta}_1)\right)\exp(\mathbf{x}_i^\top\boldsymbol{\beta}_2 - \mathbf{x}_i^\top\boldsymbol{\beta}_1).$$

Special cases of the GH structure

(i) If $\boldsymbol{\beta}_1 = \mathbf{0}$, then GH = **PH** (proportional hazards):

$$h_{\mathsf{E}}^{\mathsf{PH}}\left(t; \mathbf{x}_i\right) = h_0(t) \exp\left(\mathbf{x}_i^\top \boldsymbol{\beta}\right).$$

(ii) If $\boldsymbol{\beta}_2 = \mathbf{0}$, then GH = **AH** (accelerated hazards):

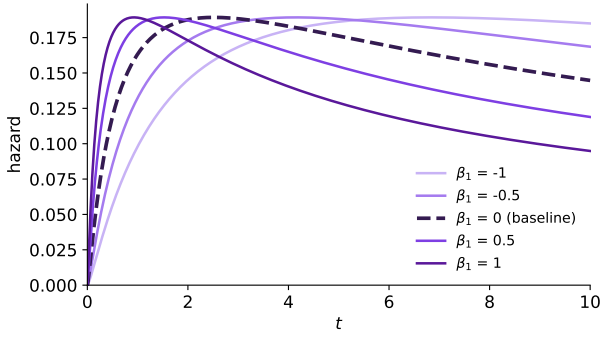
$$h_{\mathsf{E}}^{\mathsf{AH}}\left(t; \mathbf{x}_i\right) = h_0\left(t \exp(\mathbf{x}_i^\top \boldsymbol{\beta})\right).$$

(iii) If $\boldsymbol{\beta}_1 = \boldsymbol{\beta}_2$, then GH = **AFT** (accelerated failure time):

$$h_{\mathsf{E}}^{\mathsf{AFT}}\left(t; \mathbf{x}_i\right) = h_0\left(t \exp(\mathbf{x}_i^\top \boldsymbol{\beta})\right) \exp(\mathbf{x}_i^\top \boldsymbol{\beta}).$$

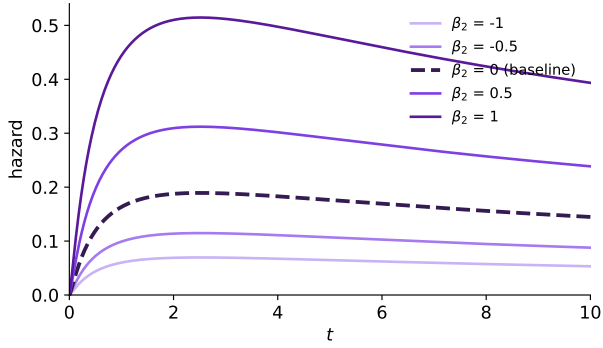
Time-dependent effects: β_1

The GH structure includes time-dependent effects through β_1, which rescales time inside the baseline hazard: $h_0\left(t\, e^{\beta_1}\right)$.



Hazard-level effects: β_2

The GH structure includes hazard-level effects through β_2, which multiplies the baseline hazard: $h_0(t)\, e^{\beta_2}$.



The baseline hazard

- For the second aim, we use the exponentiated Weibull distribution (Mudholkar, Srivastava, and Kollia 1996), a simple transformation of the Weibull distribution $F(t \mid \kappa, \sigma)$:

$$G(t \mid \kappa, \sigma, \alpha) = F(t \mid \kappa, \sigma)^{\alpha}.$$

- The corresponding hazard can be unimodal, increasing, decreasing, constant or bathtub-shaped: the "basic shapes" of the hazard function.
- The GH structure with an EW baseline hazard covers both aims. ✓
- Alternatives include the generalised gamma, the power generalised Weibull and the generalised Weibull distributions.

Maximum likelihood estimation

The likelihood function of the parameter vector $\boldsymbol{\psi}$ is

$$\begin{aligned}
\mathcal{L}_0(\boldsymbol{\psi}; \text{data}) &= \prod_{i=1}^{n} h_o(t_i; \mathbf{x}_i)^{\delta_i} S_o(t_i; \mathbf{x}_i) = \prod_{i=1}^{n} h_o(t_i; \mathbf{x}_i)^{\delta_i} \exp\left\{-H_o(t_i; \mathbf{x}_i)\right\} \\
&\overset{\text{(exercise)}}{=} \prod_{i=1}^{n} \left\{h_{\mathsf{P}}(A_i + t_i; \mathbf{z}_i) + h_{\mathsf{E}}(t_i; \mathbf{x}_i)\right\}^{\delta_i} \exp\left\{-H_{\mathsf{E}}(t_i; \mathbf{x}_i)\right\} \\
&\qquad \times \exp\left\{-\left[H_{\mathsf{P}}(A_i + t_i; \mathbf{z}_i) - H_{\mathsf{P}}(A_i; \mathbf{z}_i)\right]\right\} \\
&\propto \prod_{i=1}^{n} \left\{h_{\mathsf{P}}(A_i + t_i; \mathbf{z}_i) + h_{\mathsf{E}}(t_i; \mathbf{x}_i)\right\}^{\delta_i} \exp\left\{-H_{\mathsf{E}}(t_i; \mathbf{x}_i)\right\}.
\end{aligned}$$

The population cumulative hazard does not depend on $\boldsymbol{\psi}$, so it can be dropped: only the population hazard *rate* at death is needed.

Software and examples

- The GH model is implemented, in both the overall and relative survival frameworks, in the R package GHSurv.
- Simulating times to event from the GH structure: GHSim
- A simulated data example: GHGH
- Simulating times to event from a life table: SimLT

Application: lung cancer in women

Real data application

- Data from a population-based national cancer registry: lung cancer patients diagnosed in 2012 in the United Kingdom.
- We restricted the analysis to women with no missing data.
- We observed $n = 14{,}557$ complete cases, of whom $n_o = 12{,}138$ died before 31 December 2015.
- The median follow-up among censored patients was 3.46 years.

Covariates

- **Age at diagnosis.** Quartiles 64.9, 72.6 and 80.2 years; mean 72.0 years.
- **Tumour stage (I–IV).** Stage I: 2,434; stage II: 1,131; stage III: 3,421; stage IV: 7,751.
- **Income score** (income domain of the England Indices of Multiple Deprivation). Continuous on $(0, 1)$.
- **Comorbidity:** presence of cardiovascular disease. 4,318 patients had comorbidity indicator 1.

Model

- We use the EW baseline hazard (3 parameters).
- We fit four models for the excess hazard: PH, AFT, AH and GH.
- The likelihood is maximised using the R function `optim()`.
- We select a model using AIC.

Results: estimates (standard errors)

	PH-EW	AH-EW	AFT-EW	GH-EW
Baseline (EW)				
Scale	0.059 (0.038)	8.482 (0.724)	1.190 (0.175)	1.838 (0.374)
Shape	0.188 (0.014)	0.539 (0.046)	0.385 (0.012)	0.442 (0.033)
Power	9.175 (1.420)	1.483 (0.129)	4.387 (0.312)	3.593 (0.368)
Time scale, β_1				
Age at diagnosis	–	−0.112 (0.006)	–	0.041 (0.004)
Stage II	–	−2.977 (0.282)	–	0.691 (0.311)
Stage III	–	−6.680 (0.337)	–	1.707 (0.229)
Stage IV	–	−10.469 (0.416)	–	3.413 (0.226)
Income score	–	−2.668 (0.416)	–	0.822 (0.448)
Comorbidity	–	−1.021 (0.106)	–	0.539 (0.114)
Hazard level, β_2				
Age at diagnosis	0.022 (0.001)	–	0.032 (0.001)	0.034 (0.001)
Stage II	0.721 (0.056)	–	0.881 (0.065)	0.845 (0.069)
Stage III	1.473 (0.043)	–	1.909 (0.050)	1.849 (0.053)
Stage IV	2.211 (0.041)	–	3.003 (0.046)	3.073 (0.050)
Income score	0.527 (0.085)	–	0.744 (0.115)	0.750 (0.150)
Comorbidity	0.192 (0.021)	–	0.289 (0.029)	0.349 (0.039)
AIC	20523.141	20855.753	20189.124	**20164.911**

Reference category for stage: stage I. In the AFT model $\beta_1 = \beta_2$, so its single set of coefficients is shown once. Age and income score are centred.

Net survival

- We now calculate net survival at $t = 1, 2, 3$ and 3.9 years after diagnosis for two groups:
  1. the total population, by comorbidity (0, 1);
  2. patients aged 55–65 at stage I, by comorbidity (0, 1).
- We compare the estimates from the selected parametric model (GH-EW) with those from the nonparametric **Pohar Perme estimator** (Pohar Perme, Stare, and Estève 2012).

Net survival: total population

Net survival estimates (lower, upper):

Comorbidity	Year	GH-EW	Pohar Perme
0	1	0.407 (0.401, 0.416)	0.408 (0.398, 0.418)
	2	0.270 (0.265, 0.278)	0.268 (0.259, 0.277)
	3	0.204 (0.199, 0.211)	0.209 (0.201, 0.218)
	3.9	0.167 (0.162, 0.174)	0.182 (0.172, 0.191)
1	1	0.380 (0.371, 0.391)	0.370 (0.356, 0.385)
	2	0.254 (0.246, 0.264)	0.238 (0.225, 0.252)
	3	0.193 (0.185, 0.203)	0.169 (0.158, 0.182)
	3.9	0.158 (0.150, 0.168)	0.138 (0.126, 0.152)

Net survival: aged 55–65 at stage I

Net survival estimates (lower, upper):

Comorbidity	Year	GH-EW	Pohar Perme
0	1	0.924 (0.914, 0.935)	0.928 (0.897, 0.960)
	2	0.842 (0.827, 0.860)	0.837 (0.794, 0.883)
	3	0.770 (0.752, 0.791)	0.797 (0.749, 0.847)
	3.9	0.712 (0.692, 0.737)	0.762 (0.705, 0.823)
1	1	0.881 (0.869, 0.896)	0.901 (0.851, 0.953)
	2	0.772 (0.754, 0.793)	0.744 (0.674, 0.821)
	3	0.684 (0.662, 0.710)	0.670 (0.595, 0.755)
	3.9	0.618 (0.595, 0.647)	0.627 (0.541, 0.725)

Discussion

Discussion

- We have studied the overall and relative survival frameworks; each relies on underlying assumptions.
- Parametric survival models are useful and interpretable tools for modelling time-to-event data. It is important to understand their underlying assumptions.
- We have focused on the parametric framework, but survival functions can also be estimated with semiparametric and nonparametric approaches.

Discussion (continued)

- The relative survival framework is popular in policy-making, so it is important to have "good models" (reflect on what would make a good model).
- Other challenges: different types of censoring, informative censoring, competing risks, and cure/remission models. Survival analysis is a vast area of active research (see, e.g., Eletti et al. 2022).
- In fact ...

A challenge in the relative survival framework

- We have assumed that

$$h_o(t; \mathbf{x}) = h_P(A + t; \mathbf{z}) + h_E(t; \mathbf{x}),$$

  the sum of the hazard associated with other causes and the hazard associated with cancer.
- In some countries, life tables are stratified only by age, sex and year.
- In those countries, an extremely poor and an extremely rich patient with the same age, sex and year are assigned the same $h_P(A + t; \mathbf{z})$.

A challenge in the relative survival framework (continued)

- The same happens for a person with diabetes *vs.* a person without diabetes.
- Thus, the population hazard is either overestimated or underestimated.
- A more detailed study of this challenge can be found in Rubio et al. (2021).

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